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REVIEW

Sleep Apnea Detection Using EEG: A Systematic Review of Datasets, Methods, Challenges, and Future Directions

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Abstract

Purpose Sleep Apnea (SA) affects an estimated 936 million adults globally, posing a significant public health concern. The gold standard for diagnosing SA, polysomnography, is costly and uncomfortable. Electroencephalogram (EEG)-based SA detection is promising due to its ability to capture distinctive sleep stage-related characteristics across different sub-band frequencies. This study aims to review and analyze research from the past decade on the potential of EEG signals in SA detection and classification focusing on various deep learning and machine learning techniques, including signal decomposition, feature extraction, feature selection, and classification methodologies.

Method A systematic literature review using the preferred reporting items for systematic reviews and meta-Analysis (PRISMA) and PICO guidelines was conducted across 5 databases for publications from January 2010 to December 2024.

Results The review involved screening a total of 402 papers, with 63 selected for in-depth analysis to provide valuable insights into the application of EEG signals for SA detection. The findings underscore the potential of EEG-based methods in improving SA diagnosis.

Conclusion This study provides valuable insights, showcasing significant advancements while identifying key areas for further exploration, thereby laying a strong foundation for future research in EEG-based SA detection.

Keywords Sleep Apnea · Obstructive Sleep Apnea · Electroencephalogram · Machine Learning · Sleep Apnea Dataset · Systematic Literature review

Introduction

Sleep is a stage of unconsciousness from which a person can be aroused by sensory stimuli and is essential for healthy mental state, healthy body, normal daily activities, and learning capabilities. In other words, sleep quality directly affects the life quality. Recommended sleep duration of seven or more hours per night [1] for adults promotes active motor functions, emotional stability, concentration, sound memory, and cognitive abilities [2]. Furthermore, sleep plays a pivotal role in improving the immune system, regulating hormonal balance, including blood glucose and insulin levels, and ensuring proper functioning of major organs and physiological processes, such as the brain, heart, digestive, and respiratory systems [3]. Disrupted or insufficient sleep can lead to a myriad of health issues including cardiovascular diseases, cognitive impairments, and memory deterioration. Sleep is characterized by dynamic changes in pulse rate, breathing rate, body temperature, eye and muscle movements, and brain activity. It is classified as rapid eye movement (REM) sleep and non-rapid eye movement sleep (NREM), which cyclically alternate approximately every 90–100 min, totaling 4–6 cycles during a typical night's sleep.

The American Academy of Sleep Medicine (AASM) [4] delineates sleep stages as W, N1, N2, N3, and R (REM), with N3 representing slow-wave sleep corresponding to stages S3 and S4 of Rechtschaffen & Kales rules (R&K)

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[5]. Sleep disorders, such as sleep apnea (SA), arise from imbalanced cycling or absence of sleep stages [6]. SA affects approximately 4% of adult men and 2% of adult women [7], with underreporting due to the high cost of diagnosis and limitations of monitoring devices. A staggering 93% of middle-aged women and 82% of middle-aged men with moderate to severe SA remain undiagnosed [8]. Notably, SA is not confined to adults, with a prevalence of approximately 3% in preschool children [9]. Polysomnography (PSG), the gold standard for diagnosing sleep-related disorders, including SA, involves recording multiple physiological signals including electroencephalogram (EEG), electrooculogram, electromyogram and electrocardiogram (ECG), oronasal airflow, ribcage movements, abdomen movements, and oxygen saturation with at least eleven channels, for an entire night in a sleep laboratory setting. The PSG recordings are divided into 30-s sequence of non-overlapping time windows (segments) and then skilled neurologists or sleep experts will visually examine the signals and classify each window into one of the five sleep stages based on the predominant EEG pattern defined by AASM rules to measure the sleep quality which makes the process expensive, labor intensive [10], and uncomfortable. Moreover, due to the limited number of professionals qualified to diagnose SA in many medical facilities [11, 12], patients often face lengthy waiting lists [13]. Hence, studies focusing on the use of single biosignal are gaining interest among researchers and many works are carried out on detection or classification of SA using time and frequency analysis of heart rate variability (HRV) or non-linear method of ECG [14], pulse oximetry [15], and snoring sound analysis [16].

EEG, an important physiological signal have gained attention for SA detection and classification as it uniquely provides direct measurement of brain activity, enabling precise detection of sleep stages, arousals, and sleep quality, which are crucial for diagnosing SA. Unlike ECG, HRV, or pulse oximetry, EEG offers detailed insights into sleep architecture and disruptions making it an indispensable tool for accurate SA detection. While several reviews, such as those by [17] and [18], explore the role of EEG in diagnosing and managing sleep disorders, none specifically focus on SA. Additionally, while review articles integrating EEG with other modalities, such as functional magnetic resonance imaging (e.g., [19]), provide valuable insights, and a few reviews on SA detection exist (e.g., [20]), neither specifically prioritize methodologies that rely exclusively on EEG. To the best of author's knowledge this paper is the first systematic literature review (SLR) focusing solely on using EEG signals for potential detection of SA and classification. The major contributions of this work are as follows:

- A comprehensive examination adhering to PRISMA guidelines was undertaken in this study, encompassing publications spanning from January 2010 to December 2024.
- The primary focus of this investigation lies in exploring the viability of EEG signals in discerning, diagnosing, and categorizing SA.
- The paper identifies gaps and limitations within existing research pertaining to EEG-based SA detection and classification, thereby offering insights for prospective research endeavors.
- The insights gleaned from this review hold potential practical significance for professionals such as clinicians, researchers, and developers involved in the domain of sleep medicine.

The subsequent part of the paper is organized as follows. Section 2 presents the medical background of SA and EEG signal characteristics. Section 3 describes the methodology used in implementing SLR. Detailed discussions on the fundamental EEG processing pipeline for effective diagnosis and classification of SA are provided in Sect. 4. The paper concludes with a future roadmap in Sect. 5.

Background

SA disrupts the normal functionality of the brain by reducing oxygen levels and increasing carbon dioxide, which activates the brainstem to restore normal breathing. Key regions responsible for regulating respiration, such as the medulla, pons, and amygdala, are engaged, reflecting the profound impact SA has on neurological processes. Sleep studies provide insights into these effects, with EEG offering a direct view of brain activity changes. Alterations in EEG wave patterns across sleep stages make it a promising biosignal for SA detection and classification. This section explores the medical background and EEG signal characteristics for SA detection.

Medical Background of SA

SA is a respiratory sleep disorder that involves temporary reduction (hypopnea) or complete cessation (apnea) of breathing multiple times during night, thereby preventing deep and restful phases of sleep. These episodes may occur multiple times a night or for a short duration only during sleep, causing individuals to partially wake up multiple times during the night as they struggle to breathe. But these awakenings are typically very brief and hence, the person with SA may not even realize that they are encountering sleep disruptions. The existence and severity of SA is measured using the Apnea-Hypopnea Index (AHI) which indicates the

number of apnea events per hour of sleep [21]. Hence, identifying various sleep stages and obtaining total sleep period is necessary. An AHI less than 5 events per hour is considered normal. Some of the possible causes of SA includes anatomic abnormalities, obesity, family history/genetics, age, gender, and lifestyle habits like smoking and alcohol [22]. SA relates to cognitive issues, ischemic heart disease, cardiovascular dysfunction and stroke [23], daytime sleepiness [24] or fatigue, lack of energy, impaired memory, and can be associated with the development of several diseases, including Type-2 diabetes [25], high blood pressure, coronary heart disease, and stroke. It may also cause traffic accidents due to daytime drowsiness [23]. There are three types of SA, Obstructive SA (OSA), Central SA (CSA), and Mixed SA (MSA). An example representation of EEG epochs of normal breathing, OSA, CSA, and MSA from a subject of St. Vincent's University Hospital/University College Dublin Sleep Apnea Database (UCDDB) is shown in Fig. 1. Significant morphological differences can be seen in the signals corresponding to normal and SA cases. Fig. 1a demonstrates consistent rhythmic activity, with minor fluctuations and no abrupt changes, indicating stable neural activity typical of uninterrupted sleep. In OSA Fig. 1b, the EEG shows periods of suppressed amplitude followed by sharp spikes, reflecting arousal triggered by airflow obstruction. CSA is characterized by smoother amplitude suppression and recovery patterns, representing the absence of respiratory drive as seen in Fig. 1c. MSA in Fig. 1d combines features of both OSA and CSA, demonstrating a blend of obstructive and central mechanisms. The OSA is severe and more common causing a complete blockage of upper airways and relaxing the throat muscles to block airflow during sleep. It is identified when the frequency of obstructive respiratory events is greater than or equal to 15 events per hour. Obstructive apneas (OAs) are scored when there is at least a 90% reduction in airflow for ten or more seconds (typically scored based on the oronasal thermal flow channel) with continued respiratory muscle activity. Obstructive hypopneas (OHs) are a partial reduction in airflow, also with continued respiratory activity. Scoring OHs requires a drop in the signal by at least 30% for ten or more seconds (typically scored based on the nasal pressure transducer channel) associated with a decrease in peripheral oxygen saturation of at least 3% from baseline or an EEG arousal. Once identified, OAs and OHs are combined and averaged throughout sleep to give the AHI. OSA is further divided into 3 categories based on their

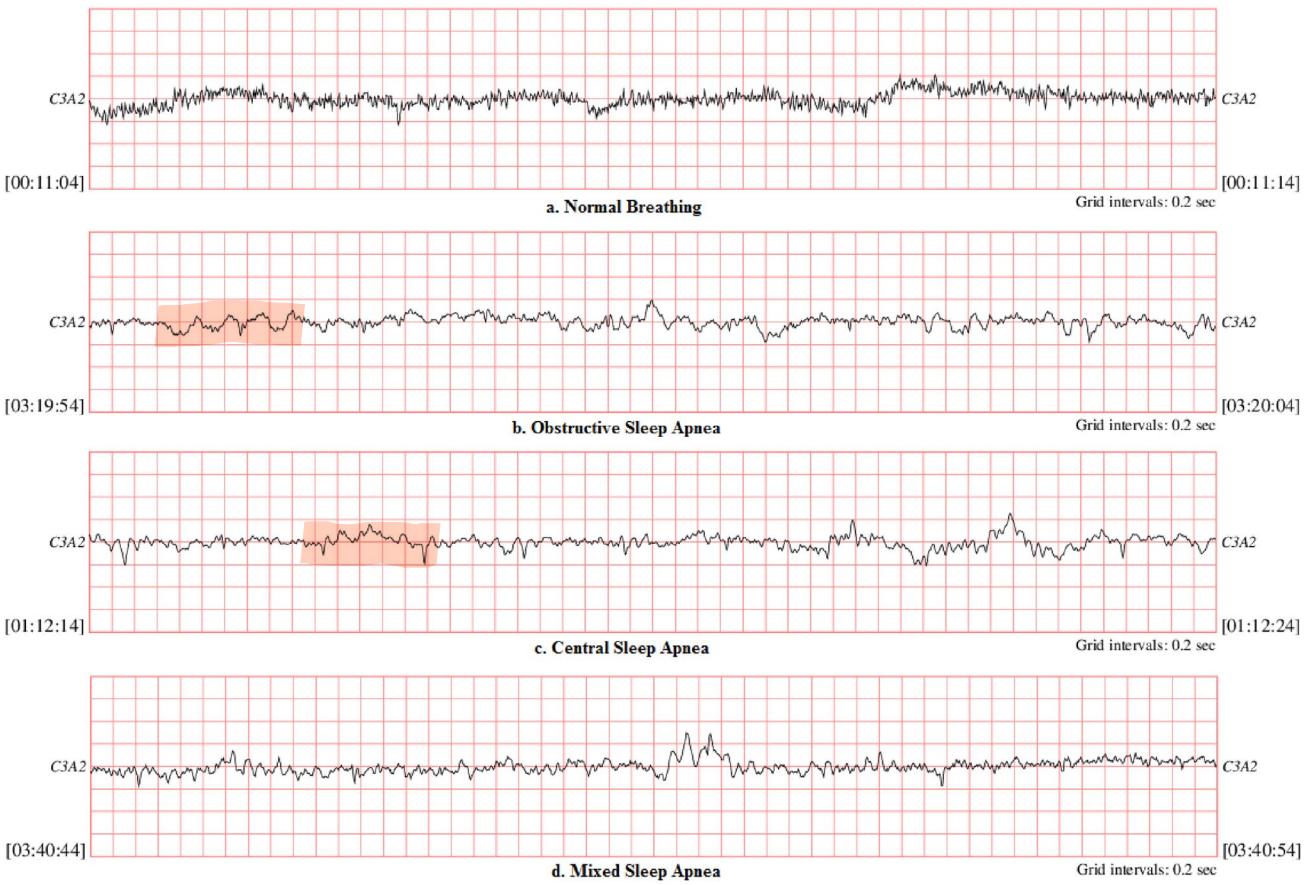


Fig. 1 EEG epochs example for normal breathing, OSA, CSA, and MSA of a subject from UCDDB database

frequencies: mild OSA (5≤AHI<15 events/hour), moderate OSA (15≤AHI<30 events/hour), and high OSA (AHI≥30 events/hour) [26]. In CSA, breathing starts and stops frequently while sleeping and the brain stops sending signals to the muscles that control breathing. It represents the state of absence of both respiratory effort of the lungs and airflow inside the nose and mouth. CSA sometimes occurs in a pattern called Cheyne-Stokes respiration where a person alternates between hyperventilating and not breathing at all. This is caused by heart failure or stroke. MSA or Complex SA is a combination of OSA and CSA, characterized by absence of respiratory effort and airflow in the first section of the event (CSA) and respiratory effort without airflow in the last section (OSA).

EEG Signal Characteristics

EEG is a non-invasive technique for recording the electrical activities of brain by placing electrodes on the scalp surface. Several channels of the EEG are recorded simultaneously from various locations on the scalp for comparative analysis of activities in different regions of the brain particularly in clinical practice. The most common EEG measurements are made with the electrodes placed in an internationally recognized 10–20 system of electrode configuration. EEG signals have amplitudes ranging from 10 to 100 μV, frequency in the range of 0.5–40 Hz, and exhibit several patterns of rhythmic or periodic activity which manifest in the form of rhythms. EEG signals are often interpreted based on the presence or absence or variations in these particular rhythms with known frequency f, usually called frequency bands, namely Delta (δ) ($0.5 \leq f < 4$ Hz), Theta (θ) ($4 \leq f < 8$ Hz), Alpha (α) ($8 \leq f < 12$ Hz), Beta (β) ($13 \leq f < 30$ Hz), and gamma (γ) ($30 \leq f < 80$ Hz). δ waves are slowest in frequency, situated at frontal cortex, found predominately in sleep stages 3 and 4 and often considered abnormal in adults who are awake. θ waves are present in temporal lobe and parietal lobe, relates to subconscious activities and appear at the beginning stages of sleep and just after awakening. α waves are the principal resting rhythm of the brain, present in parietal lobe along with occipital lobe and are common in wakeful, resting adults. These waves are strong during auditory and mental arithmetic tasks with the eyes closed and are suppressed when eyes are open. β waves are located in the frontal and parietal lobe and appears in conscious and alert state of mind.

The categorization of EEG signals based on the morphology that occurs as some events and transients as sleep spindles, K-complex, mu waves, V-waves, lambda waves, and spike waves is described in [27]. The deviation or lack of the typical expected rhythm during a particular state of the subject may suggest an anomaly. For example, in SA patients, the frequency and amplitude of sleep spindles may be altered due to repeated arousals. Also the K-complexes which are typically seen in stage 2 of sleep can be disrupted or less frequent in individuals with SA. Due to the distinguished differences in frequency content of the awake and sleep EEG, the EEG recordings are heavily used in diagnosing sleep disorders including SA.

Search Methodology

In this article, we used SLR process to identify and evaluate the available literature. SLR ensures impartial and wider scope for literature search, while also identifying research gaps. This SLR was conducted using the preferred reporting items for systematic reviews and meta-Analysis (PRISMA) approach which is a checklist to improve transparency in systematic reviews using evidence-based minimum set of items for reporting. Additionally a patient/problem, intervention, comparison (or control), and outcome (PICO) framework was used to guide the search strategy ensuring a comprehensive and focused review. The flow diagram for literature selection based on PRISMA guidelines is showed in Fig. 2.

Search Strategy

Search Terms

The search string was created using the PICO approach [28], which involves breaking down the scope of the review into four essential components: the population of interest (who the study is about), the intervention (what is being done), the comparison (what the intervention is being compared to), and the outcome (what results are being measured). This methodical decomposition, as illustrated in Table 1, ensures a comprehensive and focused search strategy that captures all relevant studies and data for the review. The following keywords were used to search these databases: `SA detection using EEG`, `SA detection EEG`, `Obstructive SA`, `mixed SA`, `central SA`, and `classification of SA`. Relevant articles were found using Search Query Language expressions for example, `("sleep apnea") OR ("OBSTRUCTIVE SLEEP APNEA") AND (EEG) AND ((Detection) OR (diognosis)) OR ("Classification of sleep apnea")`. Figure 3 presents the word clouds, where the size of each word represents its frequency or significance, for the originally extracted articles and those selected for this SLR. In the word cloud for the selected studies (Fig. 3b), the increased prominence of "EEG" suggests a strong focus on EEG-based SA detection, as opposed to the PSG signals highlighted in the original set (Fig. 3a). Additionally,

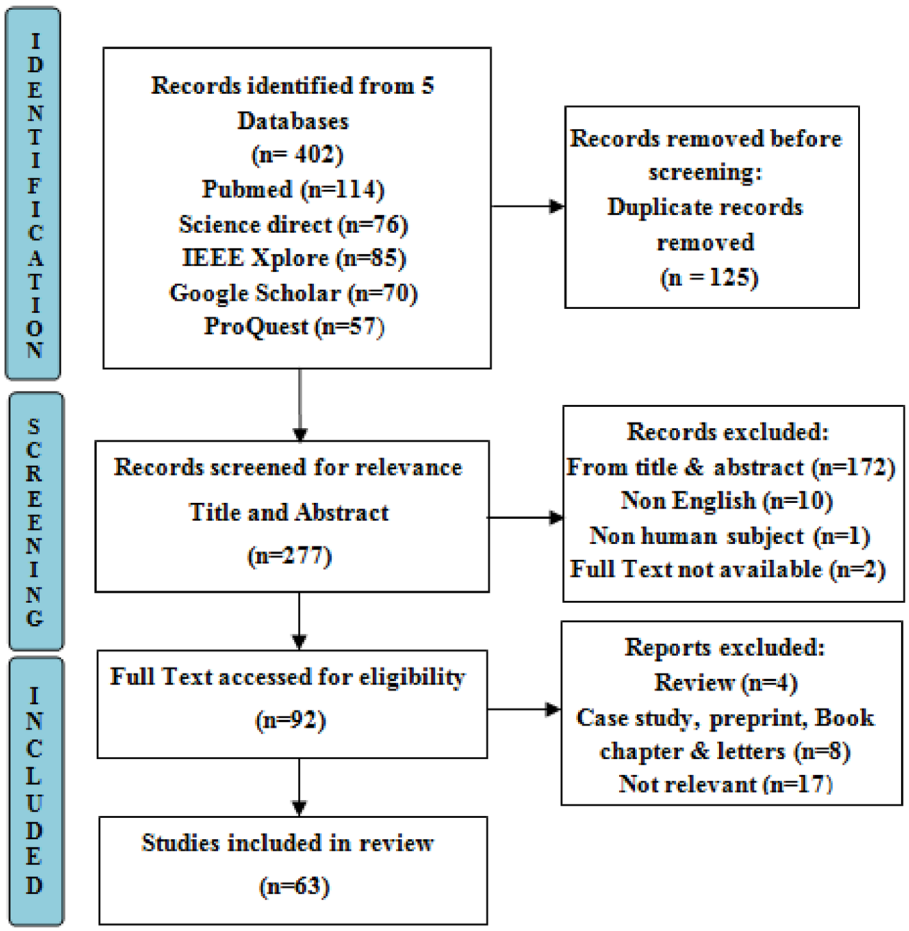


Fig. 2 PRISMA flowchart for selection of literature



Fig. 3 Word cloud of titles of **a** extracted articles and **b** selected articles

Table 1 Formation of search keywords utilizing the PICO Approach

PICO Component	Search Analysis
Population	Studies focusing on SA detection using EEG signals in adults and children
Intervention	Pre-processing, feature extraction and selection, classification methods for EEG-based SA detection
Comparison	Performance metrics of used classifiers
Outcome	Detection and classification of SA along with identification of gaps in EEG-based classification methods (ensuring that detection and classification are treated as independent categories)

the reduced visibility of the word "classification" in Fig. 3b indicates a research gap in this area.

Search Process

The literature was comprehensively sourced from five renowned international electronic databases: PubMed, Science Direct, IEEE Xplore, ProQuest, and Google Scholar, covering an expansive time period between January 2010 and December 2024. A total of 402 articles were found, specifically 114 from Pubmed, 76 on Science Direct, 85 from IEEE xplore, 57 on Proquest, and 70 from Google Scholar. Among this pool of articles, 125 were identified as duplicates and 10 were non English articles. Additionally, 172 articles were excluded based on title/abstract screening. Subsequent screening of full-text content led to the dismissal of another 17 articles due to their lack of relevance and 2 for non-availability of full text. Finally, during the data extraction phase, 13 articles were found to be unsuitable for inclusion. Consequently, a refined selection of 63 articles underwent thorough examination in their entirety.

Eligibility Criteria

Each paper identified through the keyword search was thoroughly analyzed to ensure inclusion of those that incorporated the following elements:

1. Studies focusing exclusively on the detection of SA through EEG signals
2. Papers focused on classification of SA and its various types using EEG.
3. Papers published in English language journals.
4. Papers presented in English language conference proceedings.
5. Studies involving SA detection in both adults and children.
6. Papers published within the specified time frame.

The following exclusion criteria were applied:

1. Papers published in languages other than English.
2. Preprints, short communications, letters, and review articles.
3. Studies involving non-human subjects.
4. Papers that detect SA using bio-signals other than EEG.
5. Studies that integrate EEG signals with other bio-signals, such as SpO_2 (peripheral capillary oxygen saturation), fMRI (functional magnetic resonance imaging), and ECG.
6. Papers that differentiate various sleep disorders aside from SA and/or Studies exploring the relationship between SA and other conditions such as epilepsy and blood glucose concentrations were not considered.
7. Papers published outside the specified time frame.

Fig. 4 represents the yearly distribution of the articles that met the eligibility criteria. It can be noted that significant contributions to the field of SA detection using EEG have been made since 2019, with maximum papers being published in in 2024. This surge in publications reflects the growing importance of the topic and the increasing focus on advancing research in this domain.

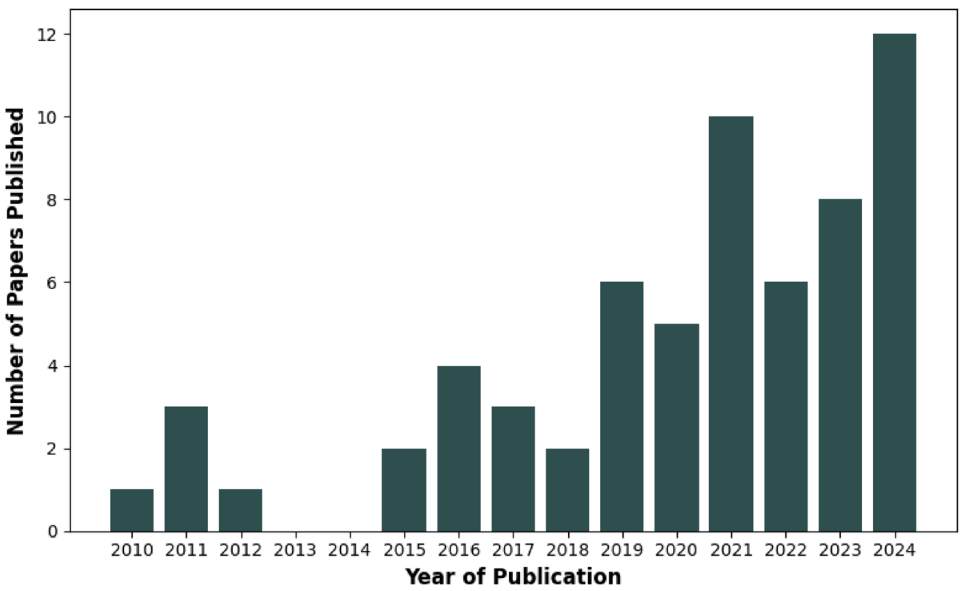


Fig. 4 Count of yearly articles that fulfilled the eligibility criteria

Screening Data Extraction and Quality Assessment

The search results retrieved from all five databases underwent consolidation, with the removal of any duplicate entries. Subsequently, two authors autonomously reviewed the titles of the search outcomes, determining the studies suitable for abstract screening. Following this initial screening, the authors independently conducted full-text evaluations and finalized the selection of studies for inclusion. Upon completion of these individual assessments, the authors convened to collate their extracted data and resolve any disparities through mutual discussions. The extracted information from eligible studies encompassed bibliometric details (such as the first author's name and publication year), the database used, type of SA detection, decomposition methodologies, techniques for feature extraction and selection, classifier types, validation techniques, and the performance metrics employed for evaluating the studies.

Discussion

In the literature review, it was found that the SA detection process using EEG involves several steps which can be considered as an EEG signal processing pipeline as illustrated in Fig. 5. The pipeline applies a set of techniques to raw EEG signals and prepare them for the subsequent and more specific analysis. The extracted data from the reviewed full text of 63 articles were systematically organized into an Excel sheet under predefined categories, such as databases used, type of SA detected, pre-processing techniques, feature extraction and selection methods, classifiers, validation techniques, and performance metrics. As we reviewed the full texts, relevant and valuable information was added to the respective categories, and gaps in the studies were identified and highlighted toward the end of the corresponding subsections. These subsections are interconnected, allowing readers to gain additional insights by cross-referencing and exploring the relationships between methodologies. This systematic and interconnected approach aligns with the principles of data triangulation, ensuring a comprehensive and robust analysis. Comprehensive summaries of the studies are presented in Tables 2, 3, 4, and 5 to enhance clarity and accessibility.

Data acquisition

Given the challenges and time limitations associated with acquiring EEG signals from individuals with SA, as well as the necessity of a sleep laboratory setup and medical expertise for signal interpretation, many researchers in the engineering field opt to utilize freely available datasets for model building purposes. This approach significantly reduces the time, cost, and effort involved in signal recording, while also facilitating the validation of research findings. Public datasets offer a standardized platform for researchers to compare their work with recent studies, thereby establishing the credibility and quality of their research. With the exception of six studies identified in this SLR, all other studies have utilized publicly available datasets. UCDDB [29] is the most widely used database in the studies followed by the MIT-BIH PSG database [30]. The details of the databases used by the studies resulting from this SLR are listed in Table 2. These datasets provide PSG signals, including EEG, recorded under standardized sleep laboratory conditions. Sampling rates typically range between 100–200 Hz, following the 10–20

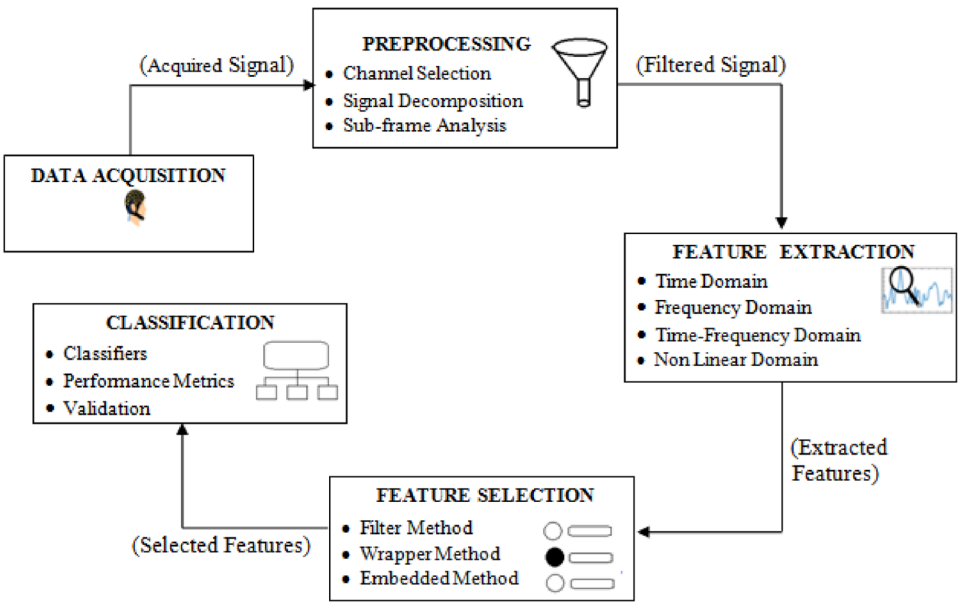


Fig. 5 Generic EEG signal processing pipeline for SA detection and classification

Table 2 Publicly available datasets used by studies of this SLR

Dataset Citations	Database	Description	Annotations	Subjects	EEG Channels	Sampling rate
[31–35] [21, 36–39] [40–47] [48–52]	UCDDB	25 full overnight polysomnograms with simultaneous three-channel Holter ECG, from adult subjects with suspected sleep-disordered breathing	Sleep stages by R&K Onset time & duration of respiratory events (obstructive, central, mixed apneas & hypopneas, & periodic breathing episodes)	25 (21 M, 4F)	C3 & C4	128Hz
[33, 37, 41, 53–56] [34, 46, 57–63] [51, 64–66]	MIT-BIH [29]	80 h of four, six, and seven-channel PSG recordings	Sleep stages and apnea	16 (All M)	C4, O2, C3	250Hz
[32, 37, 46, 66–68]	Sleep EDF	197 whole-night PSG sleep recordings, containing EEG, EOG, chin EMG, and event markers	Hypnograms (sleep patterns) manually scored according to R&K rules	197	Fpz-Cz & Pz-Oz	100 Hz
[38, 67, 69, 70]	Institute of Systems and Robotics, University of Coimbra (ISRUC) [71]	Three datagroups: (1) data of 100 subjects, one recording session per subject; (2) data from 8 subjects; two recording sessions per subject; and (3) data from one recording session of 10 healthy subjects	AASM sleep stages	G1–100 (55 M, 45F), G2–8 (6 M, 2F), G3–10 (9 M, 1F)	F3, C3, O1, F4, C4, O2	200 Hz
[33, 72–75]	Sleep Heart Health Study (SHHS) [76]	Polysomnograms were obtained in an unattended setting, usually in the homes of the participants, by trained and Certified technicians	AHI, respiratory disturbance index, sleep stages, event start and end time identification	6441 in visit 1 & 3295 in Visit-2	C3/A2 & C4/A1	125 Hz
[32]	The Wisconsin Sleep Cohort	Overnight in-laboratory ongoing longitudinal study of the causes, consequences, and history of sleep disorders, particularly SA conducted with a baseline sample of 1500 Wisconsin state employees, assessed at 4-year intervals	AHI value for each recording	1123 (608 M, 515F)	O1 and C3	100Hz
[67, 77]	Cyclic Alternating Pattern (CAP)	108 PSG recordings, each includes 3 or more EEG signals along with EOG, chin and tibial EMG, airflow, respiratory effort, SaO2, and ECG signals	Sleep stages scoring by R&K rule	2570	F3 / F4, C3 / C4, and O1 / O2	100 to 512 Hz

Table 2 (continued)

Dataset Citations	Database	Description	Annotations	Subjects	EEG Channels	Sampling rate
[78, 79]	Tianjin Chest Hospital (TCH)	Overnight PSG data recorded by the Philips Respironics Alices5 sleep monitor made in the United States	Apnea and normal events including OSA, MSA, and CSA by AASM rules	42 (25 M, 17F)	C3 and C4	100 Hz
[80]	Childhood Adenotonsillectomy Trial	Multi-center, single-blind, randomized, controlled pediatric dataset	Sleep stages & apnea/ hypopnea events by AASM	1637	C4	125 Hz
[81, 82]	Sismanoglio Amalia Fleming General Hospital of Athens (PSG Audio) [83]	Comprising 212 polysomnograms along with synchronized high-quality tracheal and ambient microphone recordings	Sleep stages and apnea events: OSA, CSA, MSA, Hypopnea, and RERA	212	C3 & C4	200 Hz
[41, 84, 85]	You snooze u win (Massachusetts General Hospital, Sleep Lab [86])	PhysioNet Challenge of various physiological signals, collected during PSG sleep studies, to detect other sources of arousal (non-apnea) during sleep	Sleep stages by AASM rules along with arousals, RERA, hypopneas, apneas (CSA, OSA, MSA)	1983 (1283 M)	F3, F4, C3, C4, O1, & O2	200 Hz

M Male; *F* Female; *RERA* Respiratory effort-related arousal

electrode placement system. The different channels and sampling rates employed by these datasets are listed in Table 2. The annotations embedded within the databases provides ground truth information that can be used to validate the effectiveness of research works and their comparison with other state-of-the-art works.

Cross-subject and Cross-dataset Validations

Most studies predominantly relied on a single database, dividing subjects for training and testing phases. Only a minority of studies such as [38, 41, 46, 67] ventured into cross-validation across more databases to assess the model's capacity for generalization across external domains. The work in [67] attempted the same methodology for SA detection on three databases and obtained improved results for the Sleep EDF database. Relying solely on one database without implementing cross-validation can restrict the applicability of test results to broader populations and settings. This advocates for future research to incorporate cross-validation with additional populations or databases to enhance the generalizability of findings. To ensure a fair and robust evaluation, it is recommended that future studies prioritize the implementation of cross-validation techniques with subject independence. An important limitation encountered in [67] was subject selection. The SA and normal subjects were directly chosen from the existing databases without any validation. Variations in SA annotation procedures across datasets posed another significant challenge in developing models that could generalize effectively across different datasets. In [33], divergent SA annotation protocols among datasets posed a notable obstacle in creating a unified Convolutional Neural Network (CNN) model applicable to both the Sleep Heart Health Study (SHHS) and UCDDB datasets, which exhibited dissimilar annotation granularity compared to the pre-segmented and pre-labeled MITBIH dataset.

Pre-processing

The real-time application of EEG is very challenging due to presence of artifacts such as eye movements/blinks [87], line noise, breathing, and muscle movements which mislead the actual cerebral activity of the recorded EEG leading to complexity in analyzing the EEG data [88]. Hence, in the pre-processing stage, the signals from EEG dataset are filtered to remove artifacts. EEG signals are also band limited to 4–40 Hz range to preserve local information about the variation patterns introduced by apnea events with respect to full-band signal. Also unique characteristics can be conserved in frequency band-limited signals for apnea detection, as opposed to those in full-band EEG signal. Infinite Impulse Response Butterworth Band pass filter (IIRB-WBPF) employing invariant technique along with pass band

and Hilbert Huang Transform (HHT)-based equiripple finite impulse response filter was used in [53] to retain the EEG sub frequencies starting from γ to δ. Their performance was compared and IIRBWBPF was found to be more efficient with increased signal to noise ratio. In [78], band pass filtering was executed using 4th order IIR filter to obtain the 4–45-Hz EEG signal and the same channel EEG signal was normalized with respect to the maximum amplitude over each epoch which can increase the generalization potential of the SA event recognition model by obtaining more robust EEG data. To prioritize other frequency components deemed more significant, the DC offset of each frame of EEG data is nullified by subtracting the mean value of the frame from each sample in the works of [21, 32, 34, 47, 50]. Subsequently, frame amplitude normalization is applied by taking maximum sample value of that frame as reference to remove unwanted amplitude fluctuations across different test frames within the same class, thereby enhancing classification performance, particularly in handling large datasets. In [44], five bandpass filters are utilized to extract band-limited EEG signals. In study [54], the EEG data are downsampled to 128 Hz. Then, a tenth-order Butterworth bandpass filter with passband frequencies ranging from 2 Hz to 45 Hz is employed to eliminate various artifacts. Additionally, a 60-Hz notch filter is incorporated to mitigate powerline interference. In study [44], an IIRBWBPF of order 20, with a passband from 0.25 Hz to 40 Hz and a sampling frequency of 128 Hz, is devised to preserve EEG sub-band frequencies ranging from δ to β bands. Lastly, in study [67], a 50-Hz notch filter is utilized to suppress unwanted AC voltage line interference from 30-second epoch raw EEG signals. In terms of signal pre-processing and enhancement, the works in [89] have used techniques to remove eye blinks and the work in [88] attempted to remove baseline wander in the EEG signals. These two being the prominent noise characteristics, must be handled carefully before subjecting the EEG data to SA analysis. The EEG signals from ISRUC database were pre-processed using a notch filter to eliminate 50-Hz electrical noise and bandpassed to retain 0.3–35 Hz using a bandpass Butterworth filter [71]. The remaining datasets listed in Table 2 contain raw EEG signals, with pre-processing, such as filtering, normalization, and artifact removal, typically left to researchers, varying based on specific application or study objectives. Pre-processing ensures the signals are band limited to 4–40Hz and are free from artifacts, noise and baseline drifts, creating a foundation for advanced analysis methods like signal decomposition and feature extraction.

Signal Decomposition

After pre-processing, EEG signals are further analyzed through decomposition into distinct frequency bands (δ, θ, α, β, and γ), each carrying distinct characteristics about brain activity during sleep. During apnea episodes, the cessation of breathing leads to an accumulation of carbon dioxide in the bloodstream. Chemoreceptors detect this change and signal the brain to awaken the individual, prompting them to resume breathing. These shifts in neural activity levels, triggered by transitions between sleep stages or from non-apnea to apnea states, induce significant variations in the EEG data across different frequency bands. Hence, following the pre-processing stage, numerous studies [21, 31, 34–38, 41–44, 46–55, 57–59, 61, 62, 62, 63, 66, 67, 72, 74, 75, 79, 82, 84, 90–96] have employed advanced techniques to decompose EEG signals into distinct frequency bands. These techniques include bandpass filtering, fast Fourier transforms (FFT), wavelet transforms, empirical mode decomposition (EMD), and variational mode decomposition (VMD). This decomposition facilitates the extraction of diverse features critical for subsequent analysis and classification. Additionally, the selection of frequency band for subsequent feature extraction plays an important role in classifier performance, as demonstrated in studies [82, 84], which highlight the impact of appropriate band selection on improving model accuracy and reliability.

Channel Selection

In the realm of EEG-based applications, the judicious selection of channels that bear physiological significance to the specific system under investigation stands as a pivotal undertaking. A subset of channels can improve performance as well as the user experience owing to the less preparation time [97]. Typically, EEG signals recorded in PSG use two channels derived from the recommended and backup leads: central (C3, C4), occipital (O1, O2), or frontal (F4, F3). Other acceptable channels are Fz-Cz, Fpz and Cz-Oz, C3 [98]. We visually represent the utilization of various channels across studies as a histogram using topoplot in Fig. 6. It is seen that many studies used the left and right central channels (C3, C4) for the detection of SA. Frontal (Fpz, F3 and F4) and Cz, Pz, Oz channels were least used. Among the 63 studies included in this SLR, 59 employ single-channel EEG recordings, while two studies, [21] and [43], averaged the data from symmetrically located pairs of EEG channels with an aim to reduce noise effects and the studies in [91] and [91] used 19-channel and 4-channel EEG data, respectively, showcasing a more comprehensive approach to capturing brain activity. The feature selection results in [78] showed that the right EEG signal (C4) has better loss, accuracy, and classification performance than the left EEG signal (C3). This suggests that during sleep, SA patients show distinct alterations in the left (C3) and right (C4) EEGs of the central area of the scalp, with the right hemisphere being more active than the left [99]. Comparison of SA detection

accuracy using features obtained in same way from one versus two symmetrical EEG channels was demonstrated in [35] which showed higher accuracy of 70.30% with combined channels - C3 and C4. Channel selection is critical for classifier performance in EEG-based applications. Studies such as [82, 84, and 85] demonstrate that model accuracy can vary significantly depending on the channels used, highlighting the need for careful channel selection to ensure robust and reliable results.

Sub-frame Level Analysis

Many studies [34, 38, 41–43, 46, 47, 50, 59, 81] [21, 32, 37, 40, 44, 45, 49, 54] employed sub-frame-based analysis for SA detection. In this approach, the test frame is divided into a shorter frame duration (1 s [32, 34, 44, 56, 82], 2 s [46], 3 s [38], 4 s [46], and 10 s [21, 31, 37, 42–44, 47, 49, 50, 54, 58, 64, 92] called sub-frame and a sensible duration of time overlap is maintained between successive sub-frames to obtain several sub-frames that can be used for training. In a N length signal with sub-frame length M, shifting by p samples with $p \ll M < N$, there will be a total $\frac{N-M}{P} + 1$ number of sub-frames. Each subframe offers a good opportunity to capture the local features more accurately. The frame length highly influences the prediction as the features around the apnea event vary significantly from the normal event [34]. The effect of variation in sub-frame length and the impact of overlapping ration in between the frames is show in the study [44]. The selection of frame length should be guided by the fundamental criterion of stationarity where the EEG signal in the time domain is segmented into small subsections (frames) such that within each frame the signal can be regarded as being approximately stationary [100]. An EEG signal is considered stationary if its statistical properties such as mean, variance, and autocorrelation, do not vary significantly over time.

In sub-frame analysis for EEG signals, stationarity ensures that features accurately represent local signal characteristics. Stationarity is a prerequisite for most feature extraction methods, as non-stationary within a frame may result in features that do not accurately represent the specific signal characteristics. If a frame is too long, the EEG signal regime may change within the frame, leading to extracted features that fail to capture localized characteristics. Shorter sub-frames, therefore, enhance the ability to capture dynamic changes in EEG patterns, particularly around apneic and non-apneic transitions. Tests such as the Augmented Dickey–Fuller test can be employed to confirm the stationarity of sub-frames. This ensures that the features extracted are robust and indicative of the underlying physiological state. Frame-based analysis offers the following advantages: (i) When features are extracted from an entire frame, the effect of changes occurring in short time intervals may not be adequately captured. Therefore, dividing the frame into sub-frames and extracting features from these sub-frames has been demonstrated to be more effective [32]. (ii) The sub-framing approach helps in minimizing the effect of errors during measurement as such errors will usually be restricted to one sub-frame and not affect the others [42]. (iii) Dividing a frame into overlapping short duration sub-frames offers an advantage of capturing local signal characteristics precisely [37]. (iv) Sub-frame-based analysis can minimize the effect of random fluctuations in the test frame. For example, an unexpected value in the original data can significantly hamper the overall analysis carried out on the entire frame. On the contrary, in the sub-frame-based analysis, only a few sub-frames will be affected by that unexpected value. Thus analysis using sub-frames is expected to obtain better characteristics in comparison to working with the entire test frame at a time [46]. (v) Sub-framing enhances the likelihood of accurately identifying specific apneic events. This is because features extracted from sub-frames display sharp changes in characteristics within an apnea frame, particularly at the transition points between apnea and non-apnea events. As the duration of the frame is typically longer than the typical duration of an apnea event, not all N samples within a frame correspond to an apneic zone. Apnea may only manifest for a limited period within the entire frame duration.

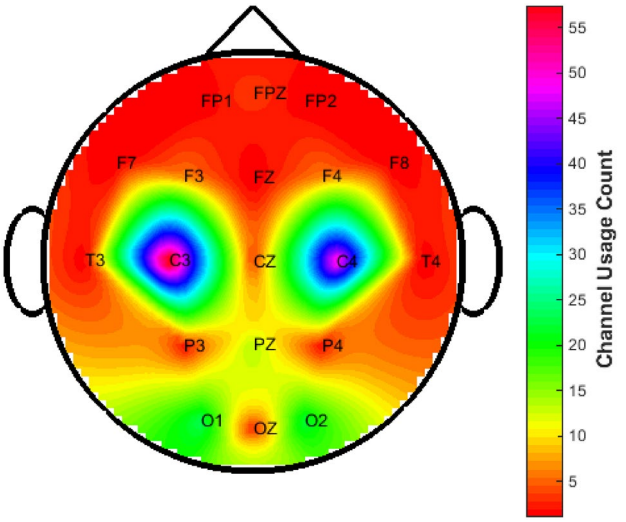


Fig. 6 Topographic plot of EEG channels illustrating their prevalence in the studies of this SLR

Feature Extraction

Detection and diagnosis of SA underscore the necessity for a suite of methods to extract pertinent features. With increasing interest from clinical experts and researchers alike, an

array of feature extraction techniques has emerged. The reviewed studies focused on feature extraction from EEG signals across four primary domains specifically in single-channel analysis: time domain, frequency domain, time-frequency domain, and non-linear analysis, as shown in Fig. 7. Time domain features, such as mean, SD, variance, skewness, kurtosis, and Hjorth parameters, capture descriptive statistics of the raw EEG, sub-bands, wavelet coefficients, or Hermite coefficients to provide detailed insights into the temporal dynamics of the EEG signals. Frequency domain analyses focus on the power distribution across EEG frequency bands using power spectral density (PSD). Other extracted metrics include range, mean absolute deviation, and interquartile range (IQR) of amplitude and frequency-modulated EEG components. These features help distinguish the spectral characteristics of apneic versus normal signals, such as increased delta activity during apnea events. Time-frequency domain techniques like wavelet transforms and EMD allow simultaneous analysis of temporal and spectral variations which are particularly effective in capturing the dynamic shifts in EEG patterns caused by apneic events. Non-linear measures such as entropy, fractal dimensions, Lempel–Ziv complexity, mutual information [102], and Petrosian fractal dimension [95] provide insights into the irregularities and complexity of EEG signals. These features highlight disruptions in brain activity associated with apnea and non-apnea transitions, offering a robust means for classification. Multi-channel analyses, while less common, primarily focus on connectivity measures such as coherence and correlation between different EEG regions. These studies examine altered functional connectivity across theta, delta, alpha, and beta bands during apnea [70]. For example, [91] utilized a 19-channel EEG setup, integrating network analysis, microstate analysis, and power spectral analysis to classify OSA. These multi-channel approaches highlight the potential of advanced connectivity analyses to provide a deeper understanding of neural interactions during apneic events. As EEG signals exhibit greater variability and complexity, the amalgamation of diverse feature extraction methods is increasingly prevalent in addressing SA. The work in [54] explored fusion of multiple domain feature parameters—Hjorth (HJ: activity, mobility, complexity), Higuchi and Katz's fractal dimension measures in time domain, rational asymmetry in frequency domain, and Renyi entropy in time-frequency domain. Table 3 highlights the various features extracted in recent studies along with the pre-processing techniques adopted. The relevance of the various features used in the studies can be found in the section 4.7. Discrete wavelet transform (DWT) [75] [48] [49] and Hilbert Transform (HT) [35] [36] [48] were widely used techniques for feature extraction followed by fully CNN [41] [34] and EMD [57] [58]. Combining multi-domain features demonstrated high potential for accurate SA detection. As can be observed from Table 3 and 5, the integration of time domain features, such as mean and variance, with non-linear features like entropy effectively captures signal variability and irregularities associated with apnea. Notably, advanced techniques like Hermite function decomposition achieved the highest accuracies. The feature extraction process for diagnosing SA becomes little complex because of the need to consider not just isolated features but also combinations thereof. It is essential to recognize that the diagnostic process does not rely solely on individual features; rather, it integrates multiple features to accurately assess the condition. The data domain, encompassing representations like time, frequency, time-frequency, and non-linear domains, along with settings such as time window length, sampling rate, channel selection, and pre-processing, shapes the granularity, relevance, and accuracy of extracted features for SA detection. These factors are critical for time-series analysis and influence feature interpretability. While Fig. 7 illustrates these domains, dataset-specific parameters like sampling rate and channel selection are detailed in Table 2, and the rationale for selecting specific time segments is discussed in Sect. 4.3.3. In the realm of the research carried out in the domain of SA detection, it is observed that numerous studies fall short in elucidating the significance of the features they extract. Consequently, it is imperative for forthcoming researchers to conscientiously address this gap in their endeavors. By doing so, they can effectively contribute to the advancement of knowledge in this field, thereby benefiting the research community.

Feature Selection

Feature extraction process is usually followed by feature selection techniques that play a pivotal role in SA detection by identifying the most significant features that contribute to distinguishing between apnea and non-apnea events. This process is essential to reduce the dimensionality of high-resolution EEG data, thereby lowering computational costs and minimizes the risk of overfitting, which would otherwise degrade classifier performance. Moreover, effective feature selection enhances the interpretability of models, allowing researchers to focus on the most informative EEG patterns, improving understanding of the physiological markers of SA. Several feature selection techniques are widely used in this context and Table 4 list those used in the studies of this SLR. Filter methods evaluate the relevance of individual features independently of classifiers using statistical metrics. These techniques are computationally efficient and provide insights into the raw discriminative power of features. However, they do not account for inter-feature dependencies. Examples of filter methods used in the studies of this SLR are One-Way Analysis of Variance (ANOVA) [35, 36, 48], Relief [48], minimum redundancy maximum relevance

Table 3 A comparative overview of the pre-processing techniques and features used in recent studies

Study	Pre-processing Techniques	Features selected	Relevance
[32]		Power spectral densities, 29 Autoregressive coefficients, and 4 statistic features (mean, SD, min and max)	Power of α and β bands was found to be different for apnea and non-apnea periods
[35]	DWT & HT	Mean, SD, skewness, kurtosis and median of γ, β, α, θ, δ and IA,IF and WIF of HT signal	Mean, max, min, SD, and variance represents events triggering awakening activity, differing between apnea and normal
[96]	HHT	Variance	Represents events triggering an awakening activity and is different between apnea and normal
[21]	FFT	Inter-band energy ratio	Offers robustness against random energy changes in frames for SA detection
[53]	DWT	Energy, Entropy, Variance	During apnea event higher-frequency bands exhibit greater relative energy compared to lower-frequency bands
[36]	DWT and HT	Mean, SD, skewness, kurtosis, and median of γ, β, α, θ, δ and IA, IF, WIF of HT signal	
[37]	BPFs	Entropy and log-variance of band-limited EEG data, mean and variance of Rician model fitted feature variation pattern	EEG variance differs between apnea and NA signals; due to its small magnitude, the logarithm of variance is used.
[101]		DFA	Scaling exponent of DFA demonstrates a notable increase during apnea
[54]	SSDM	Hjorth parameters, Renyi entropy, rational asymmetry, Higuchi, and Katz's fractal dimension measures	
[55]	TQWT	Lampel–Ziv complexity	Larger value for apnea sub-bands compared to normal
[92]		Bispectral characteristics, QPC Energy	OSA bispectrum reveals complex, highly flavored phase coupling between EEG components
[75]	DWT	SD, Power, Mean, Energy, Kurtosis, max, min, Variance, Root Mean Square	Represents events that trigger arousals, different in apnea and normal
[39]	IIRBWBPF, DWT	Band entropies	Entropy measures randomness which further grows when an apnea event occurs
[43]	BWBPF	Energy, Entropy, MAD, and kurtosis	Different between apnea and normal events
[56]	Hermite function decomposition	MIN, MAX, first, second, and third quartile, mean, SD, COV, second, third, and fourth moments of Hermite coefficients	Represents events triggering awake stage, and are different between apnea and normal
[72]		Total Power, REM Sleep Power, NREM Sleep Power for 5 bands, variance	Non-exacerbating OSA subjects has higher δ sleep power compared to the exacerbating group
[46]	VMD	Entropy and log-variance	
[57]	AEMD	Range, MAD, and IQR of FM components, COV and SD of AM component	Represents arousals, different between apnea and normal
[58]	WT, EMD	Mean absolute variation and variance of first 5 IMFs of δ band	
[34]	FFT	Energy, Geometric mean, Arithmetic mean	
[47]	FFT	Entropy	
[59]	WT	δ band power ratio	δ band power relatively decreases during the apnea events
[48]	HHT, DWT	Skewness, kurtosis, median, Hjorth parameters, ratio of the activity to the sum of the activities, entropy, max amplitude of γ, β, α, θ, δ, and IA, IF, and WIF of HT signal	
[49]	WT	Energy, Variance, Skewness, Mode, SD, Median, Geometric mean, Min, Kurtosis, Mean, Entropy, and Max of wavelet coefficients	Analyzing skewness(measure of symmetry distortion against normal distribution values of EEG segments can aid in identifying abnormal patterns

Table 3 (continued)

Study	Pre-processing Techniques	Features selected	Relevance
[61]	WT	Sleep stage, Occurrence sleep stages, Relative power of epoch	Determines sleep quality, duration of sleep stages and their transition in sleep cycle
[62]	DWT	Energy, mean, median, and SD of each wavelet coefficient	
[50]	FFT	Entropy	
[31]	IIRBWBPF	Energy, mobility, kurtosis, Skewness, IQR, and MAD for δ, θ, α, sigma, and β sub-bands	Increased IQR indicates irregularities associated with apnea events
[67]	WT	Entropy and energy of all bands	
[64]	IIRBWBPF	Energy, kurtosis, MAD, and skewness of each band	Kurtosis indicates patterns or abnormalities of apnea events by comparing epoch sample distribution with normal distribution
[74]	DWT	Minimum, Maximum, Mean, Kurtosis, Energy, Power, Variance, RMS, and SD from each of 5 sub-bands, γ, β, α, θ, and δ	
[79]	IIRBWBPF	Entropy, Variance of all bands	
[63]	IIRBWBPF	Energy, Variance of all frequency bands	
[52]	STFT	Mean spectral powers and entropy of δ, θ, α, β bands, mean, and SD of amplitude	Approximate entropy quantifies the unpredictability of fluctuations and the regularity in EEG signal that differentiate apnea and non-apnea epochs
[95]	IIRBWBPF	Mean, SD, skewness, kurtosis, PSD, approximate entropy, Largest Lyapunov exponent, Petrosian Fractal dimension of δ, theta, α bands	
[73]	DWT	minimum, maximum, variance, mean, kurtosis, rms, power for each of γ, β, α, θ, and δ sub-bands	
[94]	STFT	Shannon entropy	Measures uncertainty of EEG signals which differs during apnea
[102]		Average mutual information values and the average area of Coherence Function curves	Evaluates sleep EEG synchronization for discriminating CSA and OSA from healthy controls.
[51]	DWT	Mean, skewness, kurtosis, entropy, SD, and energy of all bands	
[93]	DWT	Max, Min, Mean, and SD of wavelet coefficients	
[90]	EMD	Power of first five data of each IMF in 5 sub-bands	
[77]		Entropy and senior Lyapunov exponents of EEG epoch	
[65]		Mean and SD of EEG epochs	
[82]	BPF	Time and spectral domain features of δ, θ, α, β, and sigma bands	
[70]	WPD	Absolute value of the decomposed α, θ, γ, β, and δ bands	Pearson correlation values were high for δ band for OSA
[91]		Power, eigenvector centrality(EC) and duration of microstate 4(MS4) of EEG sub-bands,δ, θ, α, β, and γ	Significantly increased spectral power, EC, and MS4 observed in severe OSA events

DWT Discrete Wavelet transform; *FFT* Fast Fourier Transform; *HT* Hilbert Transform, *HHT* Hilbert–Haung Transform; *IIRBWBPF* Infinite impulse response Butterworth BPF; *WT* Wavelet Transform; *IA* Instantaneous amplitudes; *IF* Instantaneous frequencies; *WIF* Weighted instantaneous frequencies; *BPF* band pass filter; *TQWT* Tunable Q wavelet transform; *VMD* variational mode decomposition; *AEMD* adaptive empirical mode decomposition; *STFT* short-term Fourier transform; *MAD* Mean absolute deviation; *IQR* interquartile range; *COV* Coefficient of variance; *SD* Standard Deviation; *DFA* Detrended fluctuation analysis; *WPD* Wavelet packet decomposition

(mRmR) [48], and Fisher Score [56]. Wrapper methods, such as Recursive Feature Elimination (RFE) used in [72, 73], Neighbor Composition Analysis (NCA) [79], Forward Selection [91], and Genetic Algorithms, iteratively selects the feature subsets in conjunction with classifiers to identify combinations that maximize predictive accuracy. While these methods are computationally intensive, they excel at capturing feature interactions. Embedded methods, such as Geometrical Separability [21, 46, 47, 58], Goodness of Feature [43], incorporate feature selection as part of the model training process, balancing accuracy, and computational efficiency. The relevance of the features is found by evaluating their utility for optimizing the learning algorithm's objective function [103]. In contrast to the other studies, the automatic classification approaches for SA events suggested in [33, 34, 41, 42, 44, 78] [66, 68, 84, 85] transferred data directly to the neural networks without requiring feature extraction or selection i.e., no manual features were used and the time-invariant properties of the original EEG signal were extracted automatically. These methods offer considerable robustness, a high level of automation, and is simple to apply to various physiological signal analysis and classification challenges increasing its convenience. However, despite these advantages, incorporating domain knowledge remains paramount. Solely relying on automatic data representations generated by deep learning models may overlook critical insights that domain expertise can provide. By focusing on selecting the most relevant features, researchers can not only improve classifier performance but also uncover clinically significant EEG biomarkers associated with SA. Further research can be directed toward exploring hybrid models that integrate time domain and frequency domain features, and leveraging advanced extraction methods to achieve greater accuracy. Utilizing advanced feature selection techniques can optimize the feature set for robust classification. Emphasizing adaptive machine learning models that dynamically optimize feature combinations during training, along with multi-feature fusion techniques, has the potential to enhance detection accuracy and reliability in clinical applications significantly.

Table 4 Feature selection techniques used in recent studies

Technique	Studies
One-way analysis of variance (ANOVA)	[35, 36, 48]
Multivariant regression analysis (MRA)	[35, 36, 48]
Fisher (F)-score	[56]
ReliefF algorithm	[48, 95]
Neighbor composition analysis (NCA)	[79]
Recursive feature elimination	[72, 73]
Minimum redundancy maximum relevance	[48]
Adaptive Feature Recalibration	[68]
Sequential feature forward selection	[91]
Geometrical Separability	[21, 37, 46, 47, 58]
Goodness of Feature	[37, 43]

Role of Classifiers

There were a diverse array of classifiers employed across the studies, ranging from single to multiple classifier models, highlighting the rich variety in machine learning and deep learning approaches within the research landscape. *k*-nearest neighbor (*k*NN) algorithm and support vector machines (SVM) emerged as the predominant classifiers used in majority of the works. In SVM, different kernels, such as linear and polynomial, RBF can be used to handle non-linear relationships of input data, while in *k*NN different distance functions can be used for classification. It is seen in [53, 56] that the SVM classifier performed best in accurately classifying the apnea and non-apnea events providing the best accuracy of 99%, while *k*NN classifier in [39] yielded accuracy of 98%. In [79], random forest algorithm (RF) has the highest classification accuracy, followed by *k*NN, and SVM has the lowest. In [46], the performance of *k*NN with different distance functions is investigated. The various classifiers used in recent studies along with their performance is listed in Table 5 and their statistical analysis is depicted in Fig. 8. In SA detection, several algorithms are employed to optimize classifier parameters for improved accuracy. Adaptive moment estimation (ADAM) has been widely adopted as an effective optimization algorithm for training neural networks in machine learning and deep learning applications [33, 41, 65, 66, 78]. The study in [37] utilized grid search to fine-tune SVM hyperparameters. Particle Swarm Optimization (PSO) was applied in [104] to optimally select Hermite functions and their dilation factors, while Bayesian optimization was leveraged in [32, 33, 54] to enhance feature extraction and classification model performance. Additionally, cross-validation is frequently employed in many studies as given in table 5 to prevent overfitting and to robustly evaluate the classifier's performance across different data subsets. Identifying the best classifier type is inherently challenging due to the diversity of testing conditions in different studies. The effectiveness of specific algorithms is highly context-dependent, shaped by the nature of the task, data attributes, and pre-processing strategies utilized. Key pre-processing steps, such as normalization and filtering, play a crucial role in enhancing classifier performance. The sampling rate is equally important, with higher rates preserving intricate EEG features, whereas lower rates can degrade spatial and temporal resolution, leading to performance loss due to missing critical information [105]. Additionally, the choice of frequency band and EEG channel significantly impacts classification accuracy, emphasizing the need to optimize

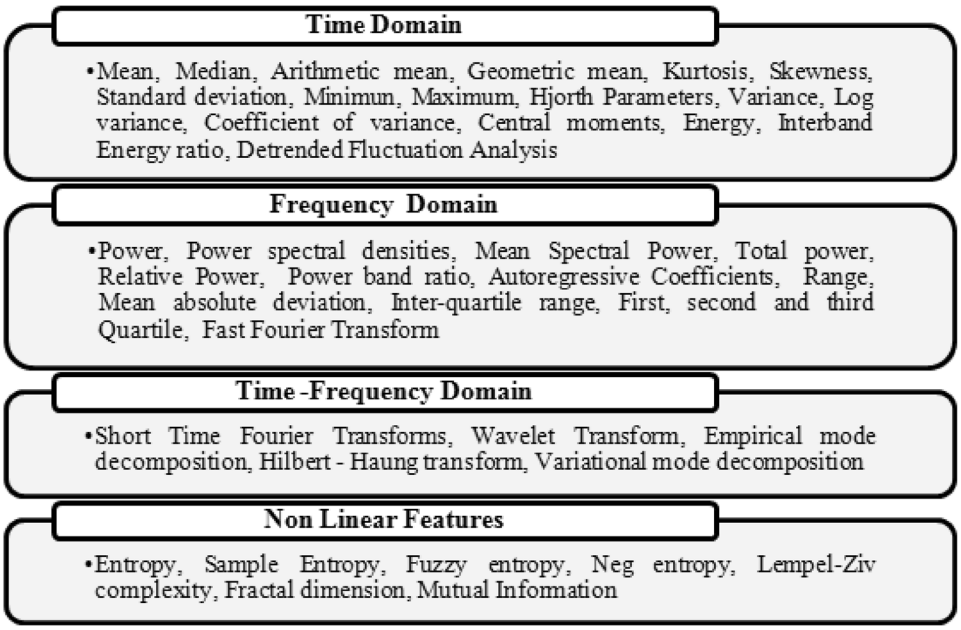


Fig. 7 Types of feature extraction methods for sleep apnea detection

both for EEG-based SA detection [82, 84]. For instance, [82] reported the highest accuracy of 82.78% in the α band using the C3-A2 channel with Naive Bayes, while [84] achieved 97.91% accuracy in the β band with the C3-M2 channel and recorded the lowest accuracy of 91.33% in the δ band with the C4-M1 channel. Furthermore, [85] highlighted that the C4-M1 channel showed slightly better performance in detecting CSA compared to C3-M2. Additionally, critical challenges like addressing data imbalance issues [106] and managing missing data must be carefully handled, especially in sensitive domains like industry and healthcare. These challenges can introduce biases into classification results, potentially leading to inaccuracies, which is particularly concerning in clinical applications where diagnostic precision is crucial. Therefore, additional research is essential to draw definitive conclusions and guide future applications effectively. Utilizing a combination of two or more models has emerged as a valuable approach for enhancing the classification of sleep data [107] [108] and may be extended for SA. This method capitalizes on the notion that individual classifiers offer unique perspectives in decision-making, and integrating multiple classifiers enables the utilization of complementary information from each. However, it is important to note that simply combining classifiers does not always lead to improved algorithm performance. The quest for an algorithm that strikes the optimal balance between performance and complexity is a particularly intriguing pursuit. Despite these advancements, further research in this field is warranted to gain a comprehensive understanding of the effectiveness of combined models in SA classification. Further exploration is warranted to identify additional nonlinear, time domain, or frequency domain features that could enhance the model's performance. Alternatively, the utilization of deep learning models could be pursued for automatic extraction of EEG signal characteristics, eliminating the need for hand-engineered features. Subsequent research endeavors could delve into investigating inter-subject variability in classifier performance among apnea patients.

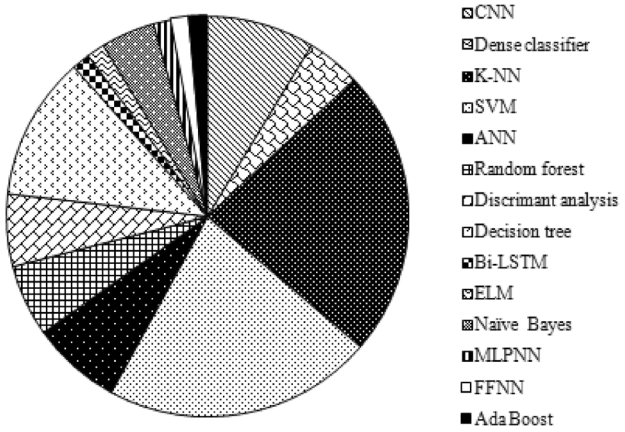


Fig. 8 Analysis of the classifiers used in the recent studies extracted from the SLR

Potential of EEG in Detecting Sleep Apnea and Its Subtypes

Recently researchers are striving to achieve real-time detection of SA solely through EEG signals, primarily due to the complexities associated with PSG and the suitable

Table 5 A comparative overview of the classifiers and their performance used in recent studies

Study	Classifiers	Performance	Validation
[31]	LSVM	UCDDB: acc−94.81, sen−93.10, spe−96.43	5-fold Cross-Validation
[32]	CNN	UCDDB: acc-80, pre−71.43, Recall 100. WSC: acc−69.15, pre−65.52, Recall−80.85	Leave-One-Out Cross-Validation
[33]	CNN	SHHS: acc−69.9, Matthew correlation coefficient−0.38	10-fold Cross-Validation
[78]	BiLSTM	TCH: acc−92.73, pre−88.46, Recall−92.47, spe−84.97, F1-score−90.42	10-fold Cross-Validation
[34]	Dense Classifier	UCDDB: acc−77.14, sen−83.9, spe−77.3	
[35]	*k*NN	UCDDB: acc−63.8, 64.3 and 70.3 for C3-A2, C4-A1 and both EEG channels	Leave 300 out cross-validation
[21]	*k*NN	UCDDB: acc−92.21, sen−90.39, spe−94.04	Subject specific
[53]	SVM, *k*NN, ANN	MITBIH SVM: acc-99, sen-100, spe-98 *k*NN: acc-75, sen-92, spe-70 ANN: acc-86, sen-100, spe-75	
[36]	FFNN	UCDDB: acc−77.27	
[37]	*k*NN	UCDDB: acc−89.22, sen−95.21, spe−83.23, GSI−0.90 MITBIH: acc−86.14, sen−91.60, spe−80.82, GSI−0.87	2,5,10 cross-validation leave one-out
[101]	SVM	acc−95.1, sen−93.2, spe−98.6	Cross-validation
[38]	EEG-MIL	UCDDB: acc−77.3, sen−80.3, spe−67.2 ISRUC: acc−74.1, sen−76.4, spe−62.5 Local hospital Data: acc−78.6, sen−81.3, spe−72.3	k-fold cross-validation
[54]	CNN, BiLSTM	MITBIH: acc−96.24, sen- 97.60, spe−96.72	Leave one-out cross-validation
[55]	DA, DT, *k*NN	MITBIH: subspace *k*NN: acc-96, sen- 95.68, spe−96.22, Matthew correlation coefficient−91.86, F1 score−0.9548	10-fold cross-validation
[92]	Multilayer NN	Private hospital data: acc−96.15	Cross-validation
[39]	*k*NN	UCDDB: acc−98.9, sen- 97.8, spe-100	
[41]	BiLSTM	UCDDB: acc−93.22, sen- 91.71, spe−93.79, F1-score−0.9276 YSYW acc−93.25, sen- 90.28, spe−91.59, F1-score−0.91 MITBIH: acc−89.41, sen- 89.24, spe−87.30, F1-score−0.90	k-fold cross-validation
[42]	CNN	UCDDB: acc−80.05, sen- 79.53, spe−80.56	
[43]	ED, SVM, *k*NN	UCDDB:ED: acc−95.10, sen- 93.20, spe−97.20, SVM: acc−93.78, sen- 91.2, spe−96.60 *k*NN: acc−92.68, sen- 91.40, spe−97.20	5-fold cross-validation
[44]	Dense Classifier	UCDDB: acc−84.9, sen−83.5, spe−78.28	
[56]	ELM, SVM	MITBIH: ELM: acc−99.53, sen- 99.47, spe−99.58 SVM: acc−99.44, sen−98.95, spe−99.83	10-fold cross-validation
[72]	QDA	SHHS: acc−79.17, sen−80.85, spe−76.00	5-fold cross-validation
[46]	*k*NN	UCDDB: acc−90.46, sen- 94, spe−86.92, GSI−0.91 MITBIH: acc−88.49, sen−95.43, spe−81.56, GSI−0.91	L eave one-out 2-,5-,10-fold cross-validation
[57]	Least-Squared SVM	MITBIH: acc−95.29, sen-100, spe−91.99	
[58]	SVM	MITBIH: acc−80.43, sen−85.59, spe−77.87	10-fold cross-validation
[34]	*k*NN	MITBIH: acc−82.28, sen−90.58, spe−77.72	Leave one-out
[60]	GreyART network	MITBIH: acc−78.10	
[93]	ANFIS	Private dataset: acc−96.25, sen−95.00, spe−97.50	
[81]	CRNN	sen−0.73, PPV−0.78	
[69]	SleepExpertNet	ISRUC: acc−0.99, Microaverage F1−0.99	
[94]	MLPNN	acc- 53.4 for quadruple classification, acc−62.6 for mild SA, acc−60.7 for moderate SA, acc−87.4 for severe SA	
[65]	Cutting edge TCCN	MITBIH: acc−0.84, pre−0.59, Recall−0.55, F1−0.57, AUC − 0.75	
[59]	*k*NN, SVM	MITBIH: SVM: acc−86.15, sen−87.50, spe−84.80 *k*NN: acc−87.03, spe−89.62, sen−84.43	2-, 5-, 10-fold cross-validation
[84]	UNETR	U snooze U win: acc−97.91	
[47]	*k*NN	UCDDB: acc−87.64, sen−89.02, spe−86.27	Leave one-out
[48]	*k*NN, SVM, ANN	UCDDB: *k*NN: acc−82.29, sen−73.44, pre−73.44 SVM: acc−83.26, sen−74.90, pre−74.9 ANN: acc−82.25, Sen−73.37, pre−73.37	32-fold
[49]	*k*NN, SVM	UCDDB: *k*NN: acc- 82.69, sen−85.92, spe-80 SVM: acc−71.15, sen:83.67, spe−65.42	
[61]	Ensemble Bagged Tree	MITBIH: acc−95.9	
[62]	SVM	MITBIH: acc-98	

Table 5 (continued)

Study	Classifiers	Performance	Validation
[50]	*k*NN	UCDDB: acc−87.64, sen−89.02, spe−86.27	
[67]	SVM, RF	SVM: ISRUC: acc-90, sen-100, spe-83, RF: ISRUC: acc-66, sen-100, spe-100 RF: sleep EDF: acc-75, sen-100, spe-100 CAP: acc-50, sen-100, spe-100	10-fold Out-of-bag
[79]	*k*NN, SVM,RF	TCH: *k*NN: acc- 85.51, recall−0.81, pre−0.82, F1−0.82, kappa−0.83 SVM: acc- 68.70, recall−0.67, pre−0.68, F1−0.67, kappa−0.54 RF: acc−88.99, recall−0.86, pre−0.89, F1−0.87, kappa−0.85	10-fold
[63]	ANN, SVM, LDA, NB	MITBIH: SVM: acc−97.14 sen−97.01, spe−97.26 LDA: acc−89.29, sen:83.82, spe−94.44 NB: acc−92.38, sen−86.45, spe−98.54 ANN: acc−94.3, sen−95.8, spe−92.6	
[64]	DT,RF	UCDDB: RF: acc−99.68 sen-100, spe−99.37	5-fold cross-validation
[73]	Extra Tree Classifier	SHHS: acc-86, pre-86, F1-86, recall-86	10-fold cross-validation
[68]	MRCNN	Sleep EDF: acc−85.8, macro F1−80.8, macro G-mean−85.7, Kappa−0.80	
[52]	SVM, LDA,NB	UCDDB:Binary classification: SVM: acc−75.90, F1−0.7587, pre−0.7691, recall−0.7590 LDA: acc−68.50, F1−0.6844, pre−0.6951, recall−0.6850 NB: acc−61.34, F1−0.6059, pre−0.6390, recall−0.6134 Multiclass classification: SVM: acc−48.24, F1−0.4646, pre−0.4815, recall−0.4824 LDA: acc−43.14, F1−0.4293, pre−0.4341, recall−0.4314 NB: acc−38.43, F1−0.3448, pre−0.3834, recall−0.3843	Leave one-out
[51]	Bagged Tree, subspace *k*NN, *k*NN, cubic SVM	MITBIH: Bagged Tree: acc−98.28, pre−0.79, recall−0.94, F1−0.85 Subspace *k*NN: acc−98.92, pre−0.86, recall−0.96, F1−0.90 *k*NN: acc−97.26, pre−0.72, recall−0.90, F1−0.79 Cubic SVM: acc−98.56, pre−0.82, recall−0.95, F1−0.87 UCDDB: Bagged Tree: acc−79.42, pre−0.41, recall−0.76, F1−0.52 Subspace *k*NN: acc−81.39, pre−0.44, recall−0.79, F1−0.56 *k*NN: acc−73.01, pre−0.31, recall−0.69, F1−0.42 Cubic SVM: acc−80.14, pre−0.42, recall−0.78, F1−0.54	10-fold cross-validation
[45]	CDNN	UCDDB: acc−71.59, sen−74.04, spe−75.44, F1−71.91	
[95]	*k*NN	acc−93.33, sen−92.30, spe−94.14, AUC − 0.98	5-fold cross-validation
[40]	EEG-CLNet	UCDDB: acc−0.74, sen−0.78, spe−0.63	5-fold cross-validation
[74]	Extra Tree ensemble	SHHS: acc−59.0, pre−59.0, recall−59.0, F1−59.0	10-fold cross-validation
[85]	CNN-GRU	U snooze u win: Overall acc−99.30 OSA: pre-100, recall−99.0, F1−99.0; CSA: pre−99.0, recall−99.0, F1−99.0	
[66]	CNN and EFCM	MITBIH: acc−99.20, NMI−93.54, ARI−94.02 Sleep EDF: acc−99.57, NMI−92.76, ARI−95.28	5-fold cross-validation
[102]	FFNN	Mean relative absolute error: 0.1450	K-fold cross-validation (K=3,5)
[82]	NB, RF, *k*NN, DT	*k*NN: acc- 80.26, F1−75.07, spe−95.78 DT: acc- 71.33, F1−71.76, spe−81.89 RF: acc- 80.18, F1−75.20, spe−96.60 NB: acc- 82.76, F1−75.21, spe−98.72	
[91]	AdaBoost	acc−88.3, spe−84.0, sen−92.0	10-fold cross-validation
[75]	SVM(RBF kernel)	SHHS: acc−0.61, pre−0.61, recall−0.61, F1−0.61	

LSVM Linear Support Vector Machine; *MRCNN* Multi-Resolution Convolutional Neural Network; *ANN* Artificial neural network; *BiLSTM* Bidirectional long short-term memory; *FFNN* Feed forward neural network; *MIL* Multiple instance learning; *DA* Discriminant analysis; *DT* Decision tree; *ED* Ensemble decision tree; *ELM* Extreme learning machine; *QDA*- Quadrature discriminant analysis; *MLPNN* Multilayer perceptron neural network; *EEG-CLNet* EEG-based collaborative learning network; *CDNN* Convolutional deep neural network; *GreyART* Grey relational analysis and Adaptive resonant theory network; *ANFIS* Adaptive neuro-fuzzy inference system; *PPV* Positive predicted value; *CRNN* Convolutional recurrent neural network; *TCNN* Temporal Convolutional Neural Network; *AUC* Area under curve; *UNETR* UNet Transformer; *GRU* Gated recurrent unit; *FFNN* Feed forward neural network; *EFCM* Enhanced Fuzzy C-Means Clustering Algorithm; *NMI* Normalized Mutual Information; *ARI* Adjusted Rand Index; *LDA* Linear discriminant analysis; *NB* Naive Bayes; *acc* accuracy; *pre* precision; *spe* specificity; *sen* sensitivity

characteristics of EEG signals for SA detection, such as the following. Firstly, EEG signals can be easily acquired through small, non-invasive electrodes placed on the scalp, making them comfortable for patients to wear during sleep. Availability of portable or wireless electrodes simplify home-based EEG acquisition without disrupting sleep or requiring visits to a sleep laboratory.

EEG is widely used in sleep disorder research due to its sensitivity in capturing electrical brain activity, essential for sleep staging [18]. Its minimal instrumentation makes EEG-based detection suitable for at-home sleep tests, presenting a promising approach to modern sleep monitoring. Advances in wearable EEG devices and consumer-level products have further enhanced accessibility.

Ear-EEG sensors, explored as an alternative to scalp EEG in [109], offer benefits like convenience and improved usability. However, they show moderate agreement with scalp EEG, with a specificity as low as 0.66 for sleep monitoring, limiting diagnostic accuracy. While promising, further advancements are needed before ear-EEG sensors can replace scalp EEG. These innovations aim to enhance the accessibility and affordability of sleep assessment and health monitoring.

Analyzing EEG signal frequency bands provides valuable insights into identifying SA. Researchers have observed differences in the power ratio of various sub-bands of EEG signals between apnea and non-apnea events. Specifically, in EEG frames with apnea events, the δ band power ratios exhibit a distinct pattern: they abruptly decrease at the onset of apnea, remain low throughout the event, and then increase again once the apnea subsides. In contrast, during non-apnea events, the δ band ratios remain relatively stable [59].

Entropy values extracted from frequency bands of EEG signals can effectively distinguish apnea signals from non-apnea [47, 50, 67]. The comparison of fuzzy entropy between normal and apnea/ hypopnea respiratory events revealed distinct patterns. Across the δ to α bands, the fuzzy entropy of apnea events was consistently lower compared to that of normal respiratory events. Conversely, in the sigma ($12 \leq f < 16$ Hz) and β bands, the trend was reversed [110].

Throughout sleep, the brain's energy predominantly resides in the low-frequency δ band, which remains dominant across all frames, regardless of whether they are apneic or non-apneic. During apnea events, when breathing pauses and the patient awakens, a sharp energy transition is observed in the high-frequency β band. While energy transitions can occur during non-apneic events due to variations in sleep stages, they are generally less pronounced compared to those seen in apnea events [34]. Figure 9 illustrates energy transients in Apnea and Non-Apnea signals derived from a subject in the MIT-BIH PSG Database. Figure 9a represents the raw time-series signals for both conditions, while Fig. 9b showcases the corresponding energy transients calculated over 60 s. As observed, energy transitions in non-apnea events exhibit smoother variations due to natural sleep stage dynamics. In contrast, apnea events display abrupt and pronounced energy shifts, indicating potential physiological disturbances associated with sleep apnea. Apnea episodes are also marked by an increased relative energy contribution of higher-frequency bands compared to lower-frequency bands [21, 41, 64, 64].

Quadratic phase coupling analysis in EEG frequency bands, utilizing bispectral density, aids in identifying OSA [92]. The bispectrum reveals complex phase coupling in OSA, providing non-Gaussian and non-linear insights beyond the power spectrum. By capturing harmonic phase relationships, it enriches signal analysis and improves clinical utility. Inter-frequency phase coherence in cortical EEG may indicate widespread synchronization driven by deeper brain structures.

In EEG signals, statistical parameters such as mean absolute values, maximum, minimum, standard deviation (SD), and variance are calculated for features derived from specific time windows or frequency bands [35–37, 48, 58, 31, 49, 51, 73, 74, 79] or on hermite coefficients (HCs) [56], amplitude and frequency-modulated components [57], or wavelet coefficients [62, 93]. These quantities help capture variations in signal amplitude and power, which are indicative of events leading to awakening activity. For instance, variance and SD reflect fluctuations in signal intensity, while maximum and minimum values highlight extreme amplitude changes during apnea events. These descriptive statistics reveal significant differences between apnea and normal conditions, emphasizing their utility in distinguishing between these states.

Apnea EEG signals are also characterized by reduced variability in frequency and amplitude components [57]. The lower range and IQR of the FM component indicate diminished instantaneous frequency (IF) variation, reflecting less dynamic neural activity. Additionally, the higher coefficient of variation (COV) but lower SD of the AM component suggest constrained amplitude dispersion. These features point to reduced complexity and impaired neural regulation during apnea episodes.

The minimum and first quartile values of HCs exhibit greater variation in apnea epochs compared to control epochs. The SD of HCs is higher in control EEG epochs, reflecting their greater variability relative to apnea epochs. Similarly, the energy of HCs is higher in control EEG epochs, indicating their larger amplitude compared to apnea epochs. Higher entropy and lower neg entropy of HCs in apnea EEG epochs highlight their increased complexity compared to control EEG epochs [56].

Analyzing high-density EEG data has unveiled a widespread reduction in EEG power across various frequency bands within a specific region overlaying the parietal cortex

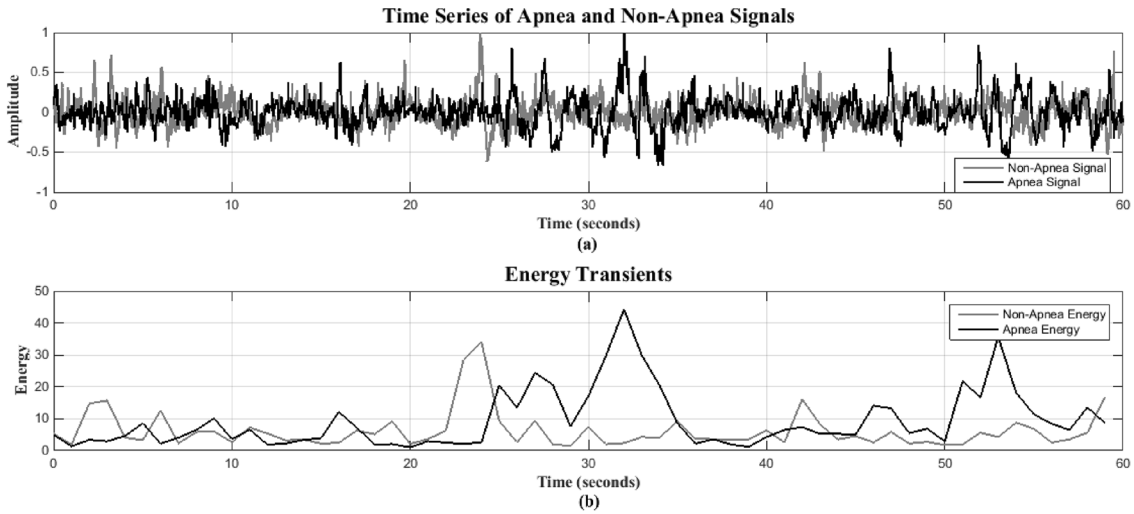


Fig. 9 Illustration of energy transients in Apnea and Non-Apnea EEG signals: **a** Time series of apnea and non-apnea signals over 60s. **b** Corresponding energy transients, highlighting sharp energy transitions during apnea compared to smoother transitions in non-apnea events, likely caused by natural sleep variations

in individuals with OSA. This reduction in neural activity in the parietal region persists across all stages and episodes of NREM sleep, indicating a consistent alteration in brain activity associated with OSA [111].

A higher value of Lempel-Ziv complexity observed in the sub-bands of apnea EEG signals, compared to those of normal EEG signals, indicates that EEG signals associated with apnea events exhibit increased randomness or complexity relative to normal EEG signals [55] with accuracy of 96%. This observation may imply reduced responsiveness of the brain in individuals with apnea, potentially due to oxygen deprivation or disrupted sleep patterns, when compared to individuals without apnea.

The scaling exponent of detrended fluctuation analysis (DFA) for apnea demonstrates a notable increase compared to normal sleep across all sleep stages [101] with accuracy reaching 95.1%. Notably, as sleep depth intensifies, so does the scaling exponents. This examination suggests that during deeper stages of sleep, the underlying dynamics of the sleep EEG tends to resemble those of Brownian noise, indicating a decrease in brain activation levels [112].

Individuals with OSA show reduced EEG connectivity, particularly in the θ, α, and β bands in the fronto-temporal regions. These decreases in functional connectivity are associated with lower sleep efficiency and higher AHI levels from the previous night's sleep, indicating a link between altered EEG connectivity patterns and OSA severity [113]. Additionally, the study in [70] found a strong correlation in brain connectivity between OSA patients and healthy individuals in the δ band, specifically in the frontal and occipital regions.

The degree of EEG synchronization between the central regions of the right and left hemispheres plays a critical role in understanding brain dysfunctions, related to SA. The average mutual information (MI) values, which quantify synchronization between brain regions, and the average area under the cross-frequency (CF) curves, reflecting interactions between different EEG frequency bands, are key non-linear and linear features, respectively, in this context [102]. During apnea, MI values decrease and CF coupling becomes abnormal, indicating disrupted neural communication, whereas in non-apnea conditions, higher MI and stable CF coupling suggest normal brain function.

The information recorded by EEG signals about sleep stages can play a significant role in apnea detection by linking apnea events to specific sleep stages. The study in [72] highlighted that the spectral characteristics of single-channel EEG, particularly when recorded from a central scalp position, show notable variations between healthy individuals and those with apnea across different sleep stages. This finding underscores the potential of leveraging EEG spectral features for the early detection and continuous monitoring of OSA. Additionally, the study in [61] demonstrated the effectiveness of incorporating sleep stage-related data for apnea detection, achieving an accuracy of 95.9% using an Ensemble Bagged Tree learner. A rare pattern identification and detection for sleep stage transitions (RAPIDEST) method was introduced in [32], defining a novel rarity score to analyze the unusualness of sleep stage sequences for detecting atypical patterns associated with OSA. The EEG segments recorded during REM sleep in [33] predicted apnea on 93% of all trials, suggesting that SA detection networks might implicitly

utilize sleep stage representations to enhance their performance. The studies in [40] and [69] explored the simultaneous detection of sleep stages and apnea, indicating a promising scope for future advancements in lightweight wearable sleep monitoring systems.

Based on these analysis of recent studies reviewed in this SLR, it is evident that EEG signals hold significant promise as a standalone biosignal for the detection of SA.

The majority of research efforts in this domain have primarily focused on discerning between apnea and non-apnea events [21, 33, 34, 39, 43–47, 50, 56, 58, 59, 75] [31, 41, 42, 53–55, 57, 60, 61, 64, 65, 67–69, 78, 81, 82, 90, 93, 94, 101], typically centering on OSA [32, 38, 40, 49, 62, 63, 72, 77, 92, 96] [31, 66, 70, 73, 74, 95]. However, there has been limited exploration into the detection of various types of SA, such as OSA, CSA, and MSA. The classification of SA events along with detection of its type into OSA and CSA is performed in [35, 36, 48, 79] [84, 85, 102], while that for all three types, CSA, MSA, and OSA, is studied only in [52, 78]. The restricted availability of mild and moderate OSA cases, along with the scarcity of CSA episodes, constitutes the primary constraints within existing databases, thereby limiting the researchers to carry out work in classification of SA. Addressing these limitations in future studies necessitates the collection of larger and more diverse datasets encompassing various types of apneas.

Forecasting Apnea

Many existing SA detection algorithms rely on analyzing several minutes of data both before and after the onset of apnea thereby restricting their suitability for real-time detection and monitoring of apnea events. Future endeavors should focus on designing and assessing machine learning and deep learning algorithms capable of forecasting SA events in advance to their occurrence in real life as a boon bestowed on humanity. Portable, single-channel EEG systems combined with deep learning classifiers like CNN-BiLSTM are promising for advancing SA detection. These systems should be lightweight, cost-effective, and suitable for at-home monitoring. Incorporating dry electrodes—non-invasive and user-friendly alternatives—could further support for future clinical, office, and home monitoring usage. However, Current studies indicate a slight drop in accuracy when dry electrodes are paired with traditional classifiers [114]. This challenge can be mitigated with advanced pre-processing techniques, highlighting the need for continued innovation. Future research should thoroughly assess and contrast the effectiveness of diverse algorithms in forecasting apnea events ahead of time paving the way for the development of personalized BCI systems. Such systems could leverage EEG signals from individual patients to deliver more reliable predictions. Given the significant health implications of untreated apnea cases, there is a critical demand for the development of AI-driven home-based apnea management systems. Such systems would not only alleviate the burden on healthcare professionals by streamlining diagnosis procedures but also offer patients relief from the intrusive data collection process typically associated with sleep labs. Moreover, they would democratize the diagnosis and monitoring of SA, making it more cost-efficient and accessible to the general public. Hence, future research efforts should persist in refining apnea classifier performance and pragmatically explore the potential benefits of real-time BCI applications in both sleep health research and clinical practice.

Challenges of SA Detection in Children

The search results revealed a greater emphasis on studies targeting the detection of SA in adults, with only one paper [80] identified that specifically investigated SA in infants, highlighting novel EEG patterns. This discrepancy can be attributed to the following distinct challenges associated with determining SA in children.

- There exist a multitude of varying etiological, diagnostic, and treatment considerations in pediatric subjects compared to adults.
- In children, there is a noted decrease in upper airway collapsibility [115], leading to fewer occurrences of respiratory events. Consequently, more strict criteria of scoring are applied for apneas and hypopneas, alongside low threshold values for the AHI used for diagnosis and grading of severity, in comparison to adults [116].
- During different stages of childhood the sleep architecture and EEG activity present substantial developmental differences [117].

Conclusion

This paper presents a comprehensive analysis of SA detection and classification utilizing EEG signals, following the rigorous PRISMA guidelines and PICO framework for SLR. By scrutinizing 63 relevant studies, the review delves into pre-processing strategies, feature extraction techniques, classifiers, and performance evaluation methodologies employed in this domain. The review underscores the diverse array of feature extraction methods utilized, ranging from time domain to frequency domain and non-linear features, reflecting the complexity of capturing SA patterns. By synthesizing these findings, it becomes evident that the selection of features significantly impacts the outcomes and effectiveness of studies. Furthermore, the study highlights the varied efficacy of classifiers, influenced by dataset characteristics and

feature representation, necessitating further investigation for robust diagnostic tools. As such researchers are encouraged to critically evaluate and justify their choice of features to enhance the credibility and reliability of their findings. The paper emphasizes the necessity for continued research endeavors to enhance the accuracy and reliability of EEG-based SA detection systems. Overall, this study contributes to advancing the understanding of EEG-based SA detection and charts a roadmap for future research directions aimed at improving diagnostic outcomes in clinical practice.

Future avenues include the exploration of novel feature extraction methods, integration of multimodal data sources for improved performance, and validation of proposed algorithms in real-world clinical scenarios. Additionally, the suggestion to conduct a meta-analysis, standardizing hardware and methodologies for fair comparison of accuracy and efficiency, offers a promising direction for future research in this field.

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